

Soheil Sayah Varg

✉ soheilsayahvarg@gmail.com • 🌐 github.com/soheilsayahvarg • 📞 +98 935 729 2456 • 🔗 linkedin.com/in/soheil-sayah

Education

Sharif University of Technology

B.Sc. in Computer Engineering

Tehran, Iran

Sep 2023 – Jul 2027

- Cumulative GPA: 19.38 / 20.0, ranked 14th of 186 students. Departmental average for this cohort: 17.24; university-wide: 15.85.
- Graduate coursework taken as an undergraduate: **Stochastic Processes** (20.0), **Reinforcement Learning**.
- Awarded 7 credits in physics on admission under the National Olympiad medalist provision.

AE High School

Diploma, Mathematics & Physics

Tehran, Iran

Sep 2020 – Jun 2023

- GPA: 19.25 / 20.0

Honors and Awards

2023: Gold Medal, Iranian National Physics Olympiad — ranked 7th nationwide among more than 10,000 participants, and 2nd nationwide in the experimental (laboratory) examination.

2023: Admitted to Sharif University of Technology through the olympiad medalist track (no entrance exam).

Research Interests

Large Language Models • Bandits • Machine Learning • ML Theory • Reinforcement Learning • RL Theory • Optimization • Probability & Statistics • Stochastic Processes

Research Experience

Data Science & Machine Learning Laboratory (DML), Sharif University of Technology

Exact Controlled Generation from Tilted Autoregressive Targets

Dec 2025 – Present

- Supervised by **Dr. Ali Rostami**. A sampling scheme that draws from a reward-tilted language-model distribution *exactly in distribution* rather than approximately, via a particle-Gibbs ladder over a batched proposal chain.
- Measured the design's assumed noise-scaling rate directly across a 32× range of sequence lengths and re-anchored it on the measured rate.
- **First-author manuscript in preparation.**

Graduate Deep RL project, team of 3, Sharif University of Technology

Model-Based Reinforcement Learning in Confounded POMDPs

Feb 2026 – Present

- Empirical implementation of Hong, Qi & Xu (ICML 2024), a theory-only paper with no released code: bridge-function identification from negative-control proxies, verified against an analytic oracle to machine precision. At full benchmark scale the **confounding-blind baseline wins in all 15 sweep cells** — estimation variance dominates the bias the method removes.
- *Role in the team of three:* read the two anchor papers and drove the implementation.

Selected Projects

Information Retrieval: Modern Information Retrieval System — Python, PyTorch, FAISS

🌐 repo

- SPLADE + dense retrieval fused by reciprocal rank fusion, cross-encoder reranking, QLoRA fine-tuning for generation.

Applied Probability: Market Predictability Study — Python, statsmodels

🌐 repo

- Re-examined my own profitable trading backtest and showed the claimed lead signal was a multiple-comparisons artifact — a coin flip reproduces its Sharpe ratio.

Optimization: Convex Optimization Algorithms — Python, NumPy, CVXPY

🌐 repo

- Chambolle-Pock and ADMM implemented from their update rules; ADMM matches CVXPY to 0.02 dB PSNR in 0.2 s.

Deep Learning: Semantic Segmentation and Deep RL — PyTorch

🌐 repo

- U-Net variants compared for segmentation; Soft Actor-Critic implemented from scratch for continuous control.

Systems: NeoGit — a version control system in C

🌐 repo

- Git-like CLI on raw POSIX file I/O: staging, commits, branching, merge, stashing, pre-commit hooks.

Teaching Experience

2024 – 2026: Teaching Assistant, Sharif University of Technology — **nine course-semesters across five terms**, including **Head TA** for Signals & Systems (Dr. Manzuri, Spring 2026), coordinating the teaching staff for a 60-student core course.

Stochastic Processes and Engineering Probability & Statistics (Dr. Najafi) • Machine Learning and Probability & Statistics (Dr.

Sharifi-Zarchi) • Data Structures & Algorithms (Dr. Abam) • Linear Algebra, twice (Dr. Rabiee & Dr. Ramezani) • Fundamentals of Programming (Dr. Rivadeh).

Aug 2023 – Present: Co-founder and physics instructor, Resonance (resonanceoly.ir) — an online Physics Olympiad school, started as a small study group, reaching students who have no selective high school near them. Teach the mechanics, electromagnetics and laboratory tracks, and write problem sets and mock exams for the national selection rounds.

2022 – Present: Physics Olympiad instructor, theory and **experimental / laboratory** tracks — core olympiad staff at **Allameh Helli 3** (NODET, Iran’s national programme for exceptional talents) and instructor at AE High School. Also introductory Python at AE Middle School and lower-secondary physics at Mahdavi School.

Academic Service and Outreach

May 2025: Juror, 13th Persian Young Naturalists’ Tournament (PYNT), Tehran — an IYPT-format research tournament run by the Ariaian Young Innovative Minds Institute under the World Federation of Physics Competitions. Judged team defenses of open-ended physics research problems.

2024 – 2025: Deputy Chairman, CEI 2024 (Computer Engineering Introduction event) • **Head of Purchase**, WSS 11 (Winter Seminar Series) • Sponsorship Staff, WSS 10.

2023 – 2025: Head of Purchase & Scientific Staff, Bugsbuzzy game development contest • Executive Staff, ICPC 2023 & 2025 • Rayan AI 2024.

Selected Coursework ([†] graduate-level)

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|-------------------------------------|--------------------------------|
| ○ Stochastic Processes [†] | ○ Probability & Statistics |
| ○ Machine Learning | ○ Computer Simulation |
| ○ Artificial Intelligence | ○ Numerical Computation |
| ○ Convex Optimization I | ○ Data Structures & Algorithms |
| ○ Linear Algebra | ○ Signals & Systems |

All courses above graded 19.5–20.0 out of 20.0.

Spring 2026: Reinforcement Learning[†], Modern Information Retrieval, Game Theory, Design of Algorithms, Database Design (*grades pending*).

Technical Skills

Languages: Python (proficient), C, C++, Java, SQL, MATLAB, $\LaTeX$

ML & Numerics: PyTorch, NumPy, SciPy, scikit-learn, statsmodels, pandas, Matplotlib, Hugging Face Transformers

Optimization & IR: CVXPY, FAISS, Gymnasium

Tools: Git, Linux, Docker, PostgreSQL, shell scripting, Proteus